

SYNTH-04: Private Synthetic Locations

Series: SYNTH — Synthetic Monitoring | **Notebook:** 4 of 6 | **Created:** December 2025 | **Last Updated:** 07/30/2026

Monitoring Internal Applications from Your Infrastructure

This notebook covers deploying and managing private synthetic locations (ActiveGates) for monitoring internal applications, APIs, and services not accessible from the public internet.

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Prerequisites

-  Access to a Dynatrace environment with Synthetic Monitoring

-  Completed SYNTH-01 through SYNTH-03
-  Infrastructure access to deploy ActiveGate (for setup)

1. Why Private Locations?

Public vs Private Locations

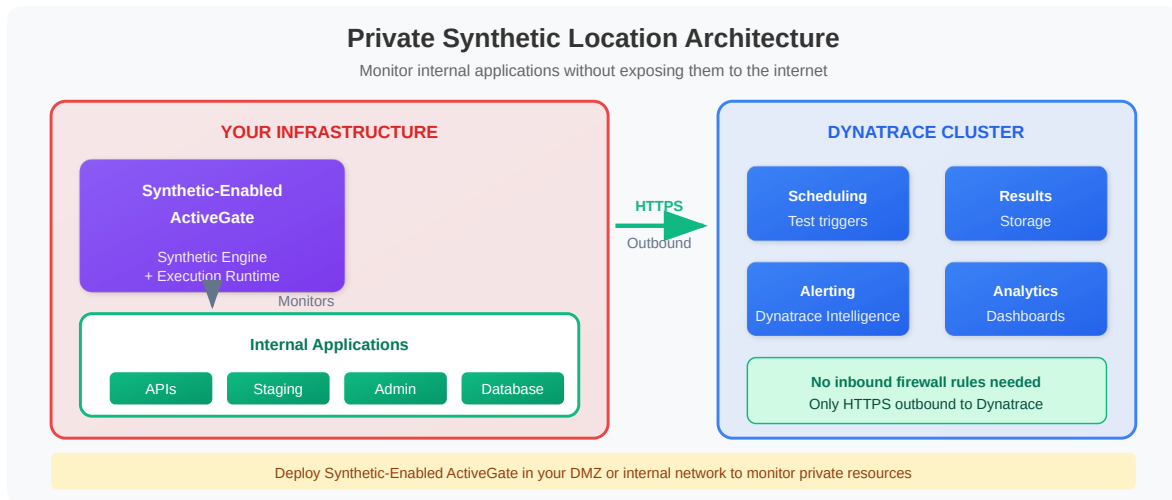
Aspect	Public Locations	Private Locations
Hosting	Dynatrace cloud	Your infrastructure
Access	Public internet only	Internal networks
Maintenance	Managed by Dynatrace	Managed by you
Security	External perspective	Behind firewall
Latency	Varies by region	Local network

Use Cases for Private Locations

Scenario	Description
Internal APIs	Services not exposed to internet
VPN-only apps	Applications requiring VPN access
Pre-production	Staging/dev environments
Security compliance	Data must not leave network
Low latency testing	Local performance baselines
Isolated networks	Air-gapped or restricted networks

2. Architecture

Private synthetic locations use ActiveGates deployed within your infrastructure to execute monitors against internal applications:



Key Points

- **Outbound only:** ActiveGate initiates all connections
- **No inbound ports:** No firewall changes for external access
- **Local execution:** Tests run inside your network
- **Results upload:** Only metrics/results sent to Dynatrace

3. Synthetic-Enabled ActiveGate

ActiveGate Capabilities

ActiveGate can serve multiple purposes:

Capability	Description
Synthetic	Private synthetic location
Metrics	Metric ingestion endpoint
Logs	Log ingestion endpoint
API	Cluster API proxy
Extensions	Extension execution

Synthetic Engine Requirements

Resource	Minimum	Recommended
CPU	2 cores	4+ cores
RAM	4 GB	8+ GB
Disk	20 GB	50+ GB
Network	HTTPS outbound	Low latency to targets
Operating system	A Linux or Windows Server release currently supported for a synthetic-enabled ActiveGate	Track the ActiveGate OS support matrix — supported releases are versioned with the ActiveGate , not with the tenant

Windows Server 2025 is supported for private synthetic locations. The OS option first appeared for synthetic-capable ActiveGates with **ActiveGate 1.341**; **SaaS 1.344** (release notes published 07/27/2026) states it in the private-location context directly: *"You can now install on Windows Server 2025 hosts for private synthetic locations."* The documentation statement is firm — published docs are live regardless of which sprint your tenant is on — but the capability ships in the **ActiveGate installer**, so confirm the ActiveGate build you are about to deploy carries it before committing a Windows Server 2025 rollout plan.

ActiveGate **1.343** (published 07/15/2026, rollout from 07/28/2026) additionally adds OS support for **Oracle Linux 9.8 and 10.2** and **Rocky Linux 9.8 and 10.2**, and flags a support-discontinuation wave running **November 2026 through January 2027** covering older Red Hat Enterprise Linux, Ubuntu 16.04, Oracle Linux, Rocky Linux, and Amazon Linux 2 builds across several architectures. Audit your private-location fleet against that list before it starts.

Browser Monitor Requirements

For browser monitors, additional requirements:

- A Chromium-family browser — **shipped with the ActiveGate**, not installed separately, so its version tracks your **ActiveGate fleet version, not your tenant version**
- Display server (X11 or headless)
- Additional RAM for browser instances

Browser baseline as of ActiveGate 1.343 (published 07/15/2026, rollout from 07/28/2026):

Browser	ActiveGate host OS
Chromium 150	Red Hat Enterprise Linux 9.7, Rocky Linux 9.8
Chrome for Testing 150	Ubuntu 20.04, 22.04, and 24.04; Amazon Linux 2023; Oracle Linux 9.7
Chrome for Testing 150 (bundled in the installer)	Windows ActiveGates

Two things follow from the browser being bundled. First, an ActiveGate that has not been upgraded is still executing clickpaths in the older browser however current the tenant is — so when the *same* clickpath behaves differently at two private locations, compare **ActiveGate versions before** suspecting the application:

```
smartscapeNodes "ACTIVEGATE"
| fields name, dt.active_gate.version, dt.active_gate.group.name, os.type,
os.version
| sort dt.active_gate.version asc, name asc
```

Second, the **1.331 floor stated below is a minimum, not a target**. It is the version that stops private locations from breaking under the security-context migration; 1.343 is the version that determines which browser your clickpaths actually run in. Both matter, for different reasons. (For the update-management model on SaaS — auto-update windows, version pinning, staged fleet upgrades — see FAQ-05.)

Breaking change (SaaS 1.343, July 2026 — staged tenant rollout from mid-July): private Synthetic locations require **ActiveGate 1.331 or newer** once 1.343 reaches your tenant, as part of migrating private-location scoping from **management zones to security context**. Upgrade any AG below 1.331 before the enforcement reaches your tenant, and review location access controls after the migration — the security-context model replaces MZ-based scoping (consistent with the platform-wide management-zone retirement; the MZ2POL series covers the broader migration). SaaS 1.343 also adds IAM-based access control for Synthetic Monitoring and API support for assigning security context to synthetic monitors.

4. Deployment Options

Option 1: Linux/Windows Installer

```
# Download from Dynatrace Hub
# Settings → Deployment status → ActiveGate

# Linux installation
sudo /bin/sh Dynatrace-ActiveGate-Linux-x86-*.sh \
  --enable-synthetic

# Verify synthetic capability
sudo systemctl status dynatracegateway
```

Option 2: Container Deployment

```
# docker-compose.yml
version: '3'
services:
  activegate:
    image: dynatrace/dynatrace-activegate:latest
    environment:
      - DT_TENANT=abc12345
      - DT_CAPABILITIES=synthetic
      - DT_API_TOKEN=${DT_API_TOKEN}
    volumes:
      - activegate-data:/var/lib/dynatrace/gateway
    ports:
      - "9999:9999"
```

Option 3: Kubernetes / OpenShift

```
# Use Dynatrace Operator with ActiveGate CRD
apiVersion: dynatrace.com/v1beta5
kind: DynaKube
metadata:
  name: dynakube
spec:
  activeGate:
    capabilities:
      - synthetic-monitoring
    resources:
      requests:
        cpu: "500m"
        memory: "512Mi"
```

 **Breaking change for "Latest Dynatrace" tenants (Sprint 1.339):** Kubernetes / OpenShift private synthetic-location pod templates now require the environment variable `METRIC_3RD_GEN_ENABLED` on the synthetic-location pod. Without it, deployments on Latest tenants fail or run without 3rd-generation metric emission. Re-pull the template from **Settings** → **Synthetic** → **Private locations** or add this env var to existing manifests before/concurrent with the Latest Dynatrace transition, and schedule it through change management.

```

// List all synthetic locations, public and private
// PREFERRED -- Smartscape. The SYNTHETIC_LOCATION node exposes location_type
DIRECTLY,
// which makes the old "identify private locations by naming convention"
workaround
// obsolete. `stage` is the lifecycle marker (GA / BETA / DEPRECATED /
COMING_SOON) --
// worth scanning for locations Dynatrace is retiring under you.
smartscapeNodes "SYNTHETIC_LOCATION"
| fields name, location_type, stage, cloud.provider, geo.country.name,
geo.city.name, id_classic
| sort location_type asc, name asc
| limit 50

// To list only private locations, add:
// | filter location_type == "PRIVATE"
//
// VERIFY THE LITERAL FIRST. location_type is confirmed present and populated,
but the
// validation tenant holds only PUBLIC locations (93 of them, no private
ones), so the
// literal "PRIVATE" is INFERRED from the field's public/private semantics
rather than
// observed. Run the summarize query in the next cell against YOUR tenant to
see the
// exact literals it returns before hard-coding one into a dashboard, alert,
or
// automation filter -- a wrong literal returns zero rows, not an error.

// FALLBACK (classic surface -- still functional, and genuinely lacks these
fields.
// This is why the previous revision of this notebook said
type/status/city/countryCode
// "are not available": on the classic entity that was true, and remains
true.)
// fetch dt.entity.synthetic_location
// | fields id, entity.name
// | sort entity.name asc
// | limit 50

```



```
// Count synthetic locations by type and lifecycle stage
// Run THIS FIRST -- it tells you the exact location_type literals your tenant
returns,
// which is what the previous cell's filter depends on.
smartscapeNodes "SYNTHETIC_LOCATION"
| summarize locations = count(), by:{location_type, stage}
| sort location_type asc, locations desc

// FALLBACK (classic surface) -- can only produce a grand total, because the
classic
// entity has no type field at all. That limitation is exactly what the old
// naming-convention workaround was compensating for.
// fetch dt.entity.synthetic_location
// | summarize {count = count()}
// | fields count
```

5. Configuration

Creating a Private Location

1. **Deploy ActiveGate** with synthetic capability
2. **Navigate to:** Settings → Synthetic → Private synthetic locations
3. **Create location:** Name, description, geographic info
4. **Assign ActiveGates:** Select which ActiveGates serve this location

Location Settings

Setting	Description
Name	Descriptive location name
Latitude/Longitude	Geographic coordinates
City/Region	Location metadata
ActiveGate nodes	Assigned ActiveGates

High Availability

For production workloads:

- Deploy 2+ ActiveGates per location
- Distribute across availability zones
- Load balancing automatic

Declarative Creation via the API

The UI flow above is the interactive path. For repeatable, reviewable, source-controlled private-location definitions, use the Environment API v2 endpoint:

```
POST /api/v2/synthetic/locations
```

API 1.344 (published 07/15/2026, **staged rollout from 07/29/2026**) adds three properties to the `PrivateSyntheticLocation` request schema — `minActiveGateCount`, `maxActiveGateCount`, and `nodeSize`. These are what make a *fully* declarative definition possible: before them, the capacity and node-sizing envelope had to be set in the UI after the location was created, so no single API call could express the finished location. Requests that omit the three properties behave exactly as they did before, so existing automation needs no change.

Property	Purpose
<code>minActiveGateCount</code>	Lower bound on ActiveGates serving the location — set it to your survivor-capacity floor (see High Availability above)
<code>maxActiveGateCount</code>	Upper bound — caps scale-out
<code>nodeSize</code>	Node sizing class for the location's ActiveGates

Because the rollout is staged, treat all three as forthcoming: send them, confirm the response accepts them, and keep the post-create UI step in your runbook until you have verified acceptance on the specific tenant you are automating.

Authentication boundary (carried over from SYNTH-01): synthetic monitor and location management still requires an **API token**, not a platform token — including through the Terraform provider (see AUTOM-04 § Authentication Boundary). Do not design a private-location automation path around platform-token auth.

Sources: [Dynatrace API release notes 1.344 \(DT docs\)](#) — changed

`PrivateSyntheticLocation` schema, added `maxActiveGateCount` ,
`minActiveGateCount` , `nodeSize` ; [Synthetic locations API \(DT docs\)](#). **Derived:** the "fully declarative" framing combines the new properties with the pre-1.344 requirement to finish sizing in the UI.

```
// Monitor executions by location
// Private locations typically have custom names (not city names like "N.
Virginia")
fetch dt.synthetic.events, from: now() - 24h
| filter endsWith(event.type, "_monitor_execution")
| summarize {
    executions = count(),
    availability_pct = round(countIf(result.state == "SUCCESS") * 100.0 /
count(), decimals: 2)
    }, by: {monitor.name, dt.entity.synthetic_location}
| fieldsAdd location = entityName(dt.entity.synthetic_location)
| sort executions desc
| limit 30
```

6. Monitoring Private Location Health

ActiveGate Health Indicators

Metric	Description	Alert Threshold
Connectivity	Connection to cluster	Any disconnect
CPU Usage	Processing load	> 80% sustained
Memory	RAM utilization	> 80%
Disk	Storage usage	> 80%
Queue Depth	Pending executions	> 100

```
// Synthetic location health status (1 = healthy)
// Surfaces location/ActiveGate availability independent of monitor results
timeseries health = avg(dt.synthetic.location.health_status),
    from: now() - 24h, interval: 1h, by: {dt.entity.synthetic_location}
```

```
// Location availability over time (last 7 days)
fetch dt.synthetic.events, from: now() - 7d
| filter endsWith(event.type, "_monitor_execution")
| makeTimeseries {
    success_count = countIf(result.state == "SUCCESS"),
    total_count = count()
}, interval: 1h, by: {dt.entity.synthetic_location}
| fieldsAdd availability_pct = success_count[] * 100.0 / total_count[]
```

```
// Execution count by location
fetch dt.synthetic.events, from: now() - 24h
| filter endsWith(event.type, "_monitor_execution")
| summarize {
    total_executions = count(),
    failed_executions = countIf(result.state == "FAIL"),
    unique_monitors = countDistinct(dt.synthetic.monitor.id)
}, by: {dt.entity.synthetic_location}
| fieldsAdd failure_rate = round((failed_executions * 100.0) /
total_executions, decimals: 2)
| fieldsAdd location = entityName(dt.entity.synthetic_location)
| sort total_executions desc
```

7. Troubleshooting

Common Issues

Issue	Symptoms	Resolution
No connectivity	Location offline	Check ActiveGate logs, network
SSL errors	Certificate failures	Install CA certs on ActiveGate
Timeouts	All tests fail	Check network routes, DNS
Resource exhaustion	Slow/failed tests	Scale ActiveGate resources
Browser issues	Clickpath failures	Check Chrome/display config

ActiveGate Logs

```
# Linux log location
/var/log/dynatrace/gateway/

# Key log files
gateway.log      # Main gateway log
synthetic.log    # Synthetic execution log
connection.log   # Cluster connectivity
```

Network Verification

```
# Test connectivity to target
curl -v https://internal-api.company.com/health

# DNS resolution
nslookup internal-api.company.com

# Test Dynatrace connectivity
curl -v https://&lt;tenant&gt;.live.dynatrace.com/api/v1/time
```

```
// Identify failing monitors by location
// To filter for specific private locations, add: | filter
dt.entity.synthetic_location == "SYNTHETIC_LOCATION-..."
fetch dt.synthetic.events, from: now() - 24h
| filter endsWith(event.type, "_monitor_execution")
| filter result.state == "FAIL"
| summarize {
    failure_count = count(),
    last_failure = max(timestamp)
}, by: {monitor.name, dt.entity.synthetic_location}
| fieldsAdd location = entityName(dt.entity.synthetic_location)
| sort failure_count desc
| limit 20
```

Summary

In this notebook, you learned:

- ✓ **Why private locations** - Internal apps, security, compliance
 - ✓ **Architecture** - ActiveGate with synthetic engine
 - ✓ **Deployment options** - Installer, container, Kubernetes (+ `METRIC_3RD_GEN_ENABLED` for Latest tenants)
 - ✓ **Configuration** - Creating and managing locations
 - ✓ **Health monitoring** - `dt.synthetic.location.health_status` metric + execution results by location
 - ✓ **Troubleshooting** - Common issues and resolution
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Next Steps

Continue to **SYNTH-05: Network Monitoring** to learn about synthetic network availability monitoring.

References

- [Private synthetic locations \(DT docs\)](#)
 - [Synthetic architecture and communication \(DT docs\)](#)
 - [ActiveGate \(DT docs\)](#)
 - [ActiveGate capabilities on Kubernetes \(DT docs\)](#)
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